Panskura Banamali College (Autonomous)

B.Sc.(H) 2nd Semester Examination , 2022

Subject : BCA **Paper: CC3P**

Paper Name :Data Structure (Practical)

F.M. : 20 **Time : 2 Hours**

A. Answer any one questions: **1x12=12**

1) Write a program in C find factorial of a number using recursion.
2) Write a program in C list of Fibonacci number using recursion.
3) Write a program in C find disk placement of Tower of Hanoi problem.
4) Write a program in C creates an array and display the array following way.

Hints:
A[0]=23
A[1]=45
A[2]=67
A[3]=43

5) Write a program in C find a number from a list of number using binary search.
6) Write a program in C sort an unsorted array using **any one** technique of followings:
   i) Bubble Sort
   ii) Insertion Sort
   iii) Quick Sort
   iv) Merge Sort

7) Write a program in C Multiplication of two arrays.
8) Write a program in C create a linked list.

B. Answer any four questions: **4x2=8**

[ONLINE Viva-Voce : Record the audio of your answers as like .mp3 format]

1. What are linear and non-linear data Structures?
2. What are the various operations that can be performed on different Data Structures?
3. What is Queue and where it can be used?
4. What is a Linked List and What are its types?
5. How to implement a queue using stack?
6. What is a Queue, how it is different from the stack and how is it implemented?
7. How is an Array different from Linked List?
8. Which data structure is used to perform recursion?
9. List the area of applications where stack data structure can be used?
10. What is a multidimensional array?

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